

Manish Singh

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SUMMARY

Senior Software Engineer specializing in Node.js/TypeScript backend systems and AI-native product development — building RAG pipelines, vector search, and LLM integrations (Vertex AI, OpenAI, MCP) alongside event-driven architecture, with a demonstrated history of leading cross-functional teams and shipping production integrations for enterprise financial clients.

WORK EXPERIENCE

Interface AI

Bengaluru, IN

Senior Software Engineer (Skills: Node.js, Python, AWS, Kubernetes, Vertex AI, PostgreSQL) Apr 2023 – Present

- Designed a **multi-agent RCA agentic workflow** (via MCP + Jira integration) that classifies issues and identifies root cause using structured LLM outputs, eliminating manual investigation for **70%** of reported issues.
- Developed the **trace-resolver cron job** to retroactively update query traces when tenants add training data; introduced a date-scoped UI filter (default: last 2 weeks) to remove stale low-confidence suggestions.
- Upgraded **Smart Discovery to v5** — built a **semantic search** pipeline generating embeddings from crawled site content and indexing them in **Pinecone**; migrated ~90% of customers to v5.
- Delivered a **RAG-based Gen AI site search** feature, orchestrating the Frontline Server and **Vertex AI** for embedding-based retrieval and content recommendations; **adopted by 80% of clients**.
- Led a team of 4 engineers and 3 QA**, mentored team members on architecture and delivery standards, coordinated **20+ client upgrade deployments**, and resolved customer CR's across the full release lifecycle.
- Implemented **silent authentication** for the OLB page with secure, encrypted data exchange; shipped **Smart Conversion** on Glia via a shared cache layer supporting cross-channel interactions live across **8+ clients**.

Interface AI

Bengaluru, IN

Software Development Engineer (Skills: Node.js, React, PostgreSQL, Docker, AWS, Jenkins) May 2022 – Mar 2023

- Integrated **FISERV core banking APIs** with supporting backend modules for data consistency and auth workflows; onboarded **13+ live clients** with 5+ recent go-lives.
- Identified and refactored sequential **REST API** calls to run concurrently using **Promise.all()**, cutting response time from **30–40s to under 10s** for 50+ enterprise clients.
- Engineered **token-authenticated conversation APIs** secured via IP whitelisting with **AWS Lambda** integration, serving **15+ clients**; added Smart Transaction support (balance checks, transfers) used by ~70% of chatbot clients.
- Integrated HealthCore APIs across multiple core systems for uptime monitoring and alerting, improving **observability for ~95% of production clients**; managed deployments via **Jenkins CI/CD** pipelines.

Twurs

Bengaluru, IN

Full Stack Developer (Skills: React.js, Next.js, GraphQL, Firebase, TailwindCSS) Aug 2021 – Apr 2022

- Replaced per-day calendar booking requests with a single aggregated monthly GraphQL query, cutting API load and improving load time by **60%**; lifted overall user engagement by **30%**.

PROJECTS

Ticketing | Node.js, Next.js, NATS Streaming, MongoDB, Kubernetes, GCP, TypeScript

Apr 2024 – Present

- Architected an event-driven microservices system with NATS Streaming pub-sub, typed events, and durable queues; published reusable NPM packages for shared utilities across services.

TECHNICAL SKILLS

- Languages & Databases:** JavaScript, TypeScript, Python, Java, SQL | PostgreSQL, MongoDB, Redis, Pinecone
- Frameworks & Libraries:** Node.js, Express, Koa, React, Next.js, TailwindCSS, Electron
- AI/LLM Engineering:** RAG Pipelines, Vector Search (Pinecone), Prompt Engineering, Vertex AI, OpenAI, MCP, Agentic Workflows
- Cloud, DevOps & Testing:** AWS, Kubernetes, Docker, Jenkins, WAF, CI/CD, Jest, Mocha
- Development Skills:** Microservices, Event-Driven Architecture, REST & GraphQL APIs, Async Programming, System Design
- Tools:** Git, Bitbucket, Jira, Kibana, VS Code, Cursor

EDUCATION

B.T.K.I.T Dwarahat

Bachelor of Technology, Computer Science & Engineering (CGPA: 8.6/10)

Dwarahat, Uttarakhand

Jul 2016 – Jul 2020